

Two-Component Extreme Value Distribution for Flood Frequency Analysis

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Theoretical considerations, supported by statistical analysis of 39 annual flood series (AFS) of Italian basins, suggest that the two-component extreme value (TCEV) distribution can be assumed as a parent flood distribution, i.e., one closely representative of the real flood experience. This distribution belongs to the family of distributions of the annual maximum of a compound Poisson process, which is a solid theoretical basis for AFS analysis. However, the two-parameter distribution of this family, obtained on the assumption of identically distributed floods, does not account for the high variability of both observed skewness and largest order statistics, so that a significant number of observed floods qualify as outliers under this distribution. The more general TCEV distribution assumes individual floods to arise from a mixture of two exponential components. Its four parameters can be estimated by the maximum likelihood method. A regionalized TCEV distribution, with parameters representative of a set of 39 Italian AFS's, was shown to closely reproduce the observed distribution of skewness and that of the largest order statistic.

INTRODUCTION

In flood frequency analysis, flood flows associated with a given recurrence interval T have been estimated from the annual flood series (AFS) using various theoretical distributions. If T is large compared with the length n of the available series, the error of the T -year flood estimate can be very large and the associated design losses quite considerable. Thus, to be acceptable for design purposes, the T -year flood estimate is required to minimize some error criterion, generally the mean square error (MSE), equaling the sum of the variance of the estimator and the square of its bias.

The choice of a flood estimator suitable for design purposes requires, as a preliminary, the identification of a "parent flood distribution," i.e., one so structured and wielding a sufficient number of parameters as to be closely representative of the real flood experience. Admittedly, that does not necessarily exhaust the problem of selecting a suitable estimator. In fact, the parent distribution, if directly used for flood quantile estimation, gives estimates that are "accurate" (small bias) but because of the limited number of available data, rather "imprecise" (large variability) and then "inefficient" (large MSE) for design purposes.

It has been argued on this score [Landwehr *et al.*, 1980; Kuczera, 1982] that preference should be given to more robust flood estimators derived from a distribution more parsimonious in the number of parameters, as long as they are more efficient (smaller MSE) and resistant over a wide parameter range. Then the loss incurred from the larger bias owing to the more parsimonious model would be offset by the saving due to the reduced estimator variance. Nevertheless, when the number of data on which the flood estimator is based is large (for example, when using regionalization techniques), the variance of the estimator will no longer be the main contribution

to its MSE. Then the most efficient estimator can be achieved by deriving it from the parent distribution.

In any case, whether the estimator is directly derived from the parent or not, the first step in flood frequency analysis must be the choice of a "good" parent flood distribution. In fact, the goodness of any suggested estimator may only be appraised and compared by sampling experiments from the parent distribution.

So far, the choice of a parent flood distribution has been based mainly on its ability to reproduce the statistical characteristics of a great number of AFS's, particular consideration being given to the behavior of the distribution right-hand tail, which more directly affects the bias of the extreme flood estimator. As the sample skewness is a statistic particularly sensitive to the behavior of the right-hand tail of a distribution, the analysis of the skewness of observed AFS's is especially useful. Within a given geographical area, AFS's exhibit a very high variability of the skewness (about the mean value for that area) unaccounted for by the most common two- or three-parameter distributions. Referred to as condition of separation [Matalas *et al.*, 1975; Landwehr *et al.*, 1978], this property of AFS skewness is connected, as is discussed below, with the presence, in some AFS's, of outliers (values significantly larger than the rest).

Two purely empirical distributions similar to one another were suggested as the flood parent able to account for the condition of separation without having to invoke spatial mixing of the distribution parameters: the Wakeby distribution suggested by H. A. Thomas Jr. and discussed by Houghton [1977, 1978] and Landwehr *et al.* [1978], and the Lambda distribution, introduced by Joiner and Rosenblatt [1971] and discussed by Stedinger [1977] and Henriques [1981].

In the present paper, to identify the parent flood distribution preference was given to theoretical-type models based on the compound Poisson process, which, if only approximately, reflects the stochastic structure of flood series. Within

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this class of models the one better apt to reproduce the upper tails of Italian AFS's is that of the maximum of a Poissonian number of a mixture of two exponential random variables. The two-component extreme value (TCEV) distribution ensuing from this approach emerges as a generalization of the Gumbel distribution.

COMPOUND POISSON PROCESSES FOR MODELING FLOOD SERIES

Some theoretical principle is needed to assist in the choice of the parent flood distribution. Because of the limited available data, besides the information from them, prior knowledge on the model structure from extreme value theory must also be used.

Asymptotic extreme value theory, applied to AFS's by E. J. Gumbel as far back as 1941, is the classical theoretical approach to flood frequency analysis. However, its basic assumption, identifying the annual flood with the maximum of a sufficiently large number of independent and identically distributed (iid) random variables, is too restrictive.

A more flexible model, and one better able to simulate physical reality, is obtained by considering the annual flood to be the maximum:

$$X = \max_{1 \leq i \leq K} Z_i \quad (1)$$

of a Poisson-distributed random number K of iid nonnegative random variables Z_i ($i = 1, 2, \dots, K$), which are also independent of K . This is tantamount to modeling X as the maximum of a compound Poisson process, as defined by Snyder [1975, p. 129]. Its cumulative distribution function (CDF) is readily obtained [e.g., Todorovic and Zelenhasic, 1970] in the form

$$F_X(x) = P[X \leq x] = \exp \{-\Lambda[1 - F_Z(x)]\} \quad x \geq 0 \quad (2)$$

where $\Lambda = E[K]$ is the parameter of the Poisson process and $F_Z(\cdot)$ is the common CDF of all Z_i , $F_Z(\cdot)$ being continuous for $x \geq 0$. Notice that CDF (2) has a discontinuity at the point $x = 0$ equal to $F_X(0) = \exp(-\Lambda)$.

The compound Poisson process has been shown by many authors [Bernier, 1967; Dauty, 1972; Zelenhasic, 1970; Todorovic and Zelenhasic, 1970] to be a valid model for the sequence of floods Z_i that exceed a given base level z_0 , provided that z_0 is large enough. Using such a sequence—the partial duration series (PDS) approach—they were able to validate CDF's of the family (2) for the largest flood exceedance in a year and then also for the annual flood X provided that $x \geq z_0$.

At low exceedance levels, neither of the two hypotheses of independence, that between events of the occurrence process or that between peak magnitudes, is any longer met. The statistical dependences then arising are accounted for by trigger-type cluster models [Cervantes et al., 1983], flood peaks occurring in clusters, each independent of one another. The generally inhomogeneous, Poisson-process characterization is now applicable to the cluster-generating process. Consequently, with the proviso of interpreting the Z_i in (1) as the largest flood peak value within a cluster, when such occur, the validity of the family of CDF's (2) in modeling an annual flood distribution may be said to extend also to low values of X .

Summing it up, CDF's of the family (2) are exact (not asymptotically derived) distributions accounting for the basic stochastic structure of series of flood peaks, more satisfactory than is the case of asymptotic extreme value distributions.

Their scope is not confined to the PDS approach, but they may confidently be applied also to the analysis of AFS's.

If Z is taken to be an exponential random variable

$$F_Z(z) = 1 - e^{-z/\theta} \quad z \geq 0 \quad (3)$$

with $\theta = E[Z]$; then (2) reduces to

$$F_X(x) = \exp(-\Lambda e^{-x/\theta}) \quad x \geq 0 \quad (4)$$

which, but for the interpretation of the parameters or range of validity, is much the same as the asymptotic extreme value type 1 (EV1), or Gumbel, distribution

$$F_X(x) = \exp\{-\exp[-(x - \varepsilon)/\theta]\} \quad (4')$$

with

$$\varepsilon = \theta \ln \Lambda \quad (5)$$

As already noted, CDF (4) has a jump at $x = 0$ where its density exhibits a "delta" component, while CDF (4') extends to negative values of x ("negative" flows) with a continuous density throughout. The magnitude of the probability jump $e^{-\Lambda}$, equaling $P[X = 0]$ in (4) and $P[X \leq 0]$ in (4'), is generally negligible, except possibly in arid or semiarid regions.

In what follows, CDF (2) will be applied to Italian data, by first considering, as is standard practice in hydrological research, the simplest CDF of this family, that given by (4), which is practically undistinguishable from the EV1 distribution. Although some of the results of such an initial analysis have already been covered in the literature [e.g., Benson, 1968], they yield useful indications for the subsequent step of selecting more suitable distributions of the same family.

SKEWNESS TEST FOR ACCEPTANCE OF THE EV1 DISTRIBUTION

Applied to AFS's, conventional goodness-of-fit tests would not seem to be powerful enough to distinguish among similarly skewed distributions nor do they reflect compatibility of a distribution with design purposes. A better test statistic would be one that emphasizes the quality of fit in the right-hand tail. The sample skewness coefficient, given for an AFS ($X_1, X_2, \dots, X_n$) by

$$G = n^{1/2} \sum_{i=1}^n (X_i - \bar{X})^3 / \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right]^{3/2} \quad (6)$$

with $\bar{X}$ denoting the sample mean, is sensitive to the goodness of the fit in the tails of a distribution.

In the case of a symmetrical distribution, G is equally sensitive to either the left- or right-hand tail. In the case of the EV1 distribution, characterized by a positive theoretical skew $\gamma = 1.14$, G is more sensitive to the right-hand tail, which makes it particularly useful as a statistical test for frequency analysis. It was already used for this purpose by Gumbel [1952], though in a somewhat approximate manner.

In carrying out such a test, we must consider that for a sample of size n , G is subject to large sampling variability and in addition is downward biased and bounded, the bounds $(-g_b, g_b)$ depending only on n , with $g_b = (n-2)/(n-1)^{1/2}$ [Kirby, 1974]. For the EV1 distribution the CDF of G , $F_G(g) = P[G \leq g]$ is dependent only on n . Matalas et al. [1975] calculated the sampling mean and standard deviation of G , $\mu(G)$, and $\sigma(G)$ for samples of size $n = 10(10)90$ by Monte Carlo simulation. The CDF of G , calculated by similar techniques, is shown in Figure 1, plotted an extremal probability paper, for sample sizes $n = 30(10)60$.

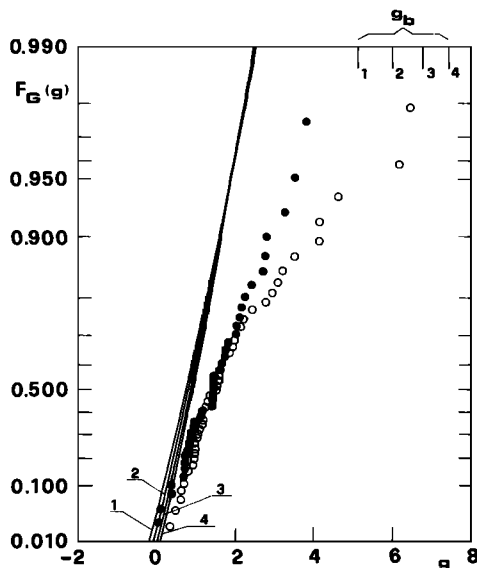


Fig. 1. Observed CDF of the skewness G for 39 Italian AFS's (solid circles) and for 46 U.S. AFS's (open circles). Sampling experiment-derived CDF and upper bound g_b of G for the EV1 distribution and for $n = 30, 40, 50, 60$ (curves and top strokes 1, 2, 3, 4, respectively).

It may be seen from Figure 1 that the acceptance interval of the skew is so large that it is very difficult to discover meaningful results by applying the skew test to a single AFS. A reliable decision about whether to accept or reject the EV1 distribution may be reached only by applying the test to a great number of AFS's and comparing observed and sampling experiment-derived CDF's of G .

TESTS FOR PRESENCE AND IDENTIFICATION OF OUTLIERS IN EV1 SERIES

Observed flood distributions often depart from the EV1 distribution for the presence of one or more outliers, i.e., observations departing so much from the remaining ones to arouse suspicion that a different mechanism is responsible for them. Then, when it is plotted on extremal probability paper, the AFS analyzed fails to show the straight line behavior expected of an EV1 variable. The actual behavior exhibited is better described by a pair of half lines, with the highest observations following the right-hand tail of a high-variability EV1 distribution [Rowe *et al.*, 1957; Potter, 1958; and others].

The sensitivity, already mentioned, of the sample skew G to the behavior of the right-hand tail of the distribution has prompted the suggestion of using it as the basis for an outlier test. Ferguson [1961] showed G to be a useful statistic for ascertaining the presence of multiple outliers in the right-hand tail of a normal sample, the critical region being defined by large values of G only [see also Hawkins, 1980, pp. 19–20]. All the more, G is applicable as an outlier test for a skewed distribution such as the EV1 distribution.

If at most one outlier is present, a preferable test [Karlin and Truax, 1960] is one based on the reduced largest order statistic

$$Y_{(n)} = (X_{(n)} - \hat{\varepsilon})/\hat{\theta} \quad (7)$$

where $X_{(n)}$ is the largest order statistic of iid EV1 variables $X_1, X_2, \dots, X_n$ and $\hat{\varepsilon}$ and $\hat{\theta}$ are maximum likelihood (ML) estimators of ε and θ . $X_{(n)}$ has to be considered an outlier if $Y_{(n)}$ is significantly large. While $(X_{(n)} - \varepsilon)/\theta$ is an EV1 variable whose

parameters, for (4'), equal $\ln n$ and 1, $Y_{(n)}$ as per (7) is not exactly an EV1, for the estimators $\hat{\varepsilon}$ and $\hat{\theta}$ are themselves random variables. The CDF of $Y_{(n)}$, $F_{Y_{(n)}}(y) = P[Y_{(n)} \leq y]$ is not available in closed form and was evaluated numerically by generating 20,000 samples of size $n = 30(10)50$ from an EV1 distribution with $\varepsilon = 0$ and $\theta = 1$, estimating ε and θ from each sample by the ML method, and calculating $Y_{(n)}$ for each sample from (7). Such sampling experiment CDF's are shown in Figure 2. Limited sampling experiments with different values of ε and θ were also carried out. They showed that the CDF of $Y_{(n)}$ is practically independent of ε and θ .

The statistic $Y_{(n)}$ merely tests whether the largest order statistic is or is not an outlier, while the skewness G tests the presence of one or more outliers without identifying their number. In the presence of more than one outlier, recourse may be made to a heuristically justified, iterative procedure by which extreme observations are successively rejected until neither the G nor the $Y_{(n)}$ test, applied to the remaining observations, gives significant values. It is referred to as the "forward selection rule" [Hawkins, 1980, p. 63].

THE ITALIAN ANNUAL FLOOD SERIES: CONDITION OF SEPARATION AND PRESENCE OF OUTLIERS

Thirty-nine stream-gauging stations in central and southern Italy, operating on streams altogether lacking in regulation or diversions and for which the longest annual instantaneous flood series were available (34 to 49 years with an average of 40 years), were selected among those operated by Servizio Idrografico Italiano. The drainage area comprised between 34 and 16,545 km², but only in four cases was it less than 100 km² and only in three cases larger than 3000 km². Floods in this region are generated by storm rainfall, the influence of snowmelt being negligible.

For each series the EV1 parameters ε and θ were estimated by the ML method. By using such estimates, the reduced order statistic $Y_{(n)}$ was calculated, together with the skewness coefficient G . The observed CDF's of G and $Y_{(n)}$ are plotted in Figures 1 and 2, respectively, superimposed on the corre-

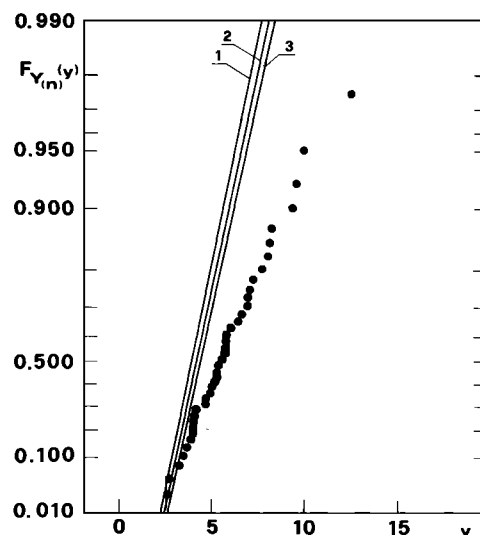


Fig. 2. Observed CDF of the reduced n -year flood $Y_{(n)}$ for 39 Italian AFS's (solid circles). Sampling experiment-derived CDF of $Y_{(n)}$ for the EV1 distribution and for $n = 30, 40, 50$ (curves 1, 2, 3, respectively).

sponding CDF obtained by the sampling experiments described above. The former CDF's are seen to separate from the latter more and more sharply as the values of G and $Y_{(n)}$ increase. Thus the condition of separation under the EV1 distribution is seen to hold also for Italian streams.

For both tests a significant number of series, namely 12 out of 39, exhibit critically large values at the 0.05 significance level. It is noteworthy that the series with significantly high skewness are also those with significantly high $Y_{(n)}$ values. The observed high variability of G and $Y_{(n)}$ is strictly related to the presence, in some series, of outliers. To make sure that outliers were not attributable to large measurement errors, daily rainfall measurements carried out in or close to the basin at the time of the event were analyzed. The corresponding daily rainfall data were always found to be associated with a large return period and, in most cases, were themselves outliers in the annual maximum rainfall EV1 distribution. Furthermore, it was ascertained from plots on extremal paper that the classing of an AFS value as an outlier did not depend on whether distribution (4) or (4') was considered. The following admittedly coarse simple experiment pointed out to the outliers being the main cause for the arising of the condition of separation. By applying to the 12 critical Italian AFS's the forward selection rule described above, eight of them revealed the presence of a single outlier, three exhibited two outliers, and just one series exhibited four. Once outliers were eliminated in this way, the CDF of G calculated with the remaining observations was found to agree very closely with the theoretical CDF of G obtained by data generation from the EV1 distribution. No separation was any longer apparent.

In this respect, the flood data of Italian basins display similar statistical characteristics to flood data in the United States. The similarity may be seen from Figure 1, where the observed CDF of G for the 46 U.S. series (with $n = 60$) used by Houghton [1977] and Henriques [1981] is also plotted. The disparity between U.S. and Italian data for high values of G may be attributed to relaxation of the upper bound on G with increasing n .

POWER LAW TRANSFORMATIONS OF THE EV1 DISTRIBUTION

Two basic mechanisms can give rise to AFS's that apparently have outliers: (1) Floods come from a more heavily tailed distribution than the one assumed and (2) floods do not all arise from the same distribution. Within the family of distributions whose CDF is given by (2), more heavily tailed distributions than EV1's are obtained by considering a parametric family of power law transformations of the flood magnitude Z of the type suggested by Box and Cox [1964], given by

$$\begin{aligned} Z_v &= h_v(Z) = (Z^v - 1)/v & v \neq 0 \\ Z_v &= h_v(Z) = \ln Z & v = 0 \end{aligned} \quad (8)$$

and looking for a value of the parameter v such that the observed distribution of the transformed variable Z closely approximates that of an exponential random variable. Then for the distribution of Z we can assume, instead of (3), the CDF

$$F_Z(z) = 1 - e^{-h_v(z)/\theta} \quad z \geq 1 \quad v \geq 0 \quad (9)$$

and for the annual flood X , instead of (4), the CDF

$$F_X(x) = \exp[-\Lambda e^{-h_v(x)/\theta}] \quad x \geq 1 \quad v \geq 0 \quad (10)$$

the EV1-distributed variable being now

$$X_v = \max_{1 \leq i \leq k} h_v(Z_i) = h_v(X)$$

Note that CDF (10), in a similar manner to CDF (4), exhibits a discontinuity. Thus X may be said to be EV1 distributed only on condition that the magnitude of the jump, here given by $F_X(1) = \exp(-\Lambda)$, be negligible.

Among the various transformations that one may consider for the same purpose, transformations (8) are of particular interest in that the resulting power extreme value (PEV) distributions (CDF's (10)) include both the two-parameter extreme value type 2 (EV2) and type 1 (EV1) distribution as particular cases for $v = 0$ and 1, respectively. For v values within these limits the smaller v , the more heavily tailed (and the larger the skewness of) the annual flood distribution. Thus the PEV distribution, which exactly—not merely asymptotically—characterizes the distribution of the maximum in a Poissonian sequence of iid random variables, scores better than the classical extreme value distribution in two respects: It is physically more plausible and is suitable to fit data covering a wider range.

In order to ascertain whether separation in the Italian AFS's could be accounted for by the PEV distribution for a suitable value for v , the mean $\bar{g}_v$ and the standard deviation $s(g_v)$ of the sample skew g_v of the transformed data X_v of the 39 AFS's here considered were calculated for various values of v . They are plotted as a function of v in Figure 3. Both $\bar{g}_v$ and $s(g_v)$ are found to increase monotonically with v , each with different values of the gradient. For the sake of comparison in the same figure are also shown the theoretical mean $\mu(G)$ and standard deviation $\sigma(G)$ of the sampling skew of the EV1 distribution, calculated by data generation techniques for a sample size $n = 40$.

A possible criterion in selecting the best value of v for use with the set of the 39 Italian AFS's is that of inducing equality between observed and theoretical mean skewness, $\bar{g}_v = \mu(G)$, so that on the average the upper flood quantiles are neither overestimated or underestimated. Within the set of PEV distributions the one with such a value of v should provide the best model for flood upper quantile estimation. By itself, however, the condition $\bar{g}_v = \mu(G)$ does not suffice for the distri-

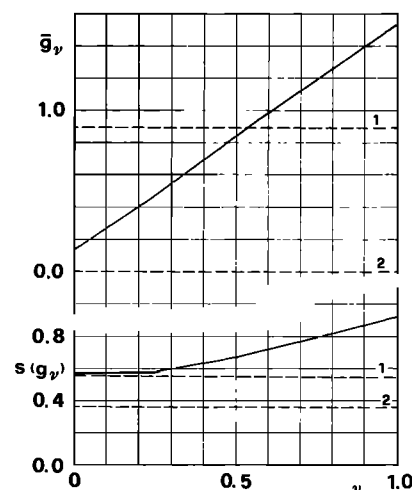


Fig. 3. Observed mean $\bar{g}_v$ and standard deviation $s(g_v)$ of the skew versus v for v power transformed data. Sampling experiment-derived mean and standard deviation of the skew for the EV1 (dashed lines 1) and the normal (dashed lines 2) distributions and for $n = 40$.

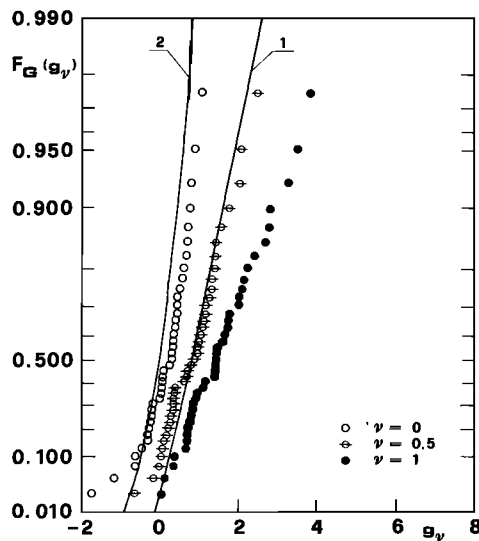


Fig. 4. Observed CDF of the skew G for ν power transformed data ($\nu = 0, 0.5, 1$). Sampling experiment-derived CDF of G for the EV1 (curve 1) and the normal (curve 2) distributions and for $n = 40$.

bution to be acceptable as a good parent flood distribution. To eliminate the condition of separation, as defined by *Matalas et al.* [1975], we must further ascertain that the same value of ν also satisfies the condition of equality between observed and theoretical standard deviation of the skew.

For the Italian AFS's, as can be seen from Figure 3, it is found that $\bar{g}_\nu$ does not equal $\mu(G)$ for the same value of ν for which $s(g_\nu)$ equals $\sigma(G)$. For $\nu = 0.5$ we have, very nearly, $\bar{g}_\nu = \mu(G)$, but the observed standard deviation of the skew is too large: $s(g_\nu) > \sigma(G)$. This is brought out also in Figure 4, which displays the observed CDF of g_ν for Italian data under the transformation (8) for $\nu = 0, 0.5$, and 1, together with the theoretical CDF of G obtained by data generation for the EV1 distribution and $n = 40$ (curve 1). The observed CDF of g_ν for $\nu = 1$, i.e., for the EV1 distribution, lies systematically to the right of the theoretical CDF (curve 1), while the observed CDF for $\nu = 0$, i.e., for the EV2 distribution, lies systematically to the left. Thus the historical data are systematically more skew than samples generated from the EV1 distribution and less skew than samples generated from the EV2 distri-

bution. Therefore, similar to what was shown by *Shen et al.* [1980] for U.S. data, the EV1 and EV2 distributions give lower and upper bounds, respectively, for extreme event predictions also for Italian data.

The PEV distribution for $\nu = 0.5$ provides a markedly better fit than either the EV1 or EV2 distribution. It does not, however, fully account for the observed skewness variability. To ascertain the significance of the disparity, a chi-square goodness of fit test was carried out, assuming the data to be independently generated from a $\nu = 0.5$ PEV distribution (implying the absence of distortion from spatial correlation). In the test, equal-probability classes were used, the interval division points being obtained from the CDF of G , calculated by data generation for a sample size n varying between 34 and 49 years. The number of degrees of freedom was identified with the number of classes minus two to allow for the fact that the power parameter ν is estimated from the condition $\bar{g}_\nu = \mu(G)$. Table 1 shows that the hypothesis is rejected at the 5% level of significance and just passes at the 1% level. The observed skew was too small in a large number of series. By plotting such transformed AFS's on extremal paper, it was seen by eye that the low skew was associated with overestimation of the upper quantiles, whereas the influence of low flows and the truncation point was negligible.

Thus statistical evidence arouses doubts against accepting the PEV distribution with one and the same regional value of ν as a good parent distribution. Furthermore, even allowing for the confounding effect of large sampling uncertainty, no noticeable spatial trend in ν or any relation between ν and drainage area was apparent, nor was it possible to recognize any subdivision into smaller regions characterizable by different ν values.

Observed and theoretical CDF's of the skewness were also compared on the assumption of a lognormal distribution, which is one frequently used in flood frequency analysis and specifically suggested for the interpretation of Italian AFS's by *Cicioni et al.* [1973]. The theoretical CDF of G calculated by *Snedecor and Cochran* [1967] for the normal distribution is plotted as curve 2 in Figure 4. The appropriate comparison is with the observed CDF of g_ν for $\nu = 0$, which embodies the required log scaling. The comparison shows that the skewness behavior of Italian AFS's is not fully accounted by the lognormal distribution either. In fact, although its mean value, which

TABLE 1. Chi-Square Goodness of Fit Test for skewness of PEV for $\nu = 0.5$, LN, and Regionalized TCEV Model

Expected Number of Observations E	PEV Model for $\nu = 0.5$; Chi-Square Statistic = 14.31*; $f = 5$; $\alpha = 0.0138^\dagger$			LN Model; Chi-Square Statistic = 19.70*; $f = 5$; $\alpha = 0.0014^\dagger$			TCEV Model; Chi-Square Statistic = 4.97*1 $f = 4$; $\alpha = 0.2900^\dagger$	
	Range of Interval, Upper Limit of G		Number of Observations of g_ν for $\nu = 0.5$, O	Range of Interval, Upper Limit of G		Number of Observations of g_ν for $\nu = 0$, O	Interval Upper-Limit of G for $n = 40$	Number of Observations of g_ν for $\nu = 1$, O
	From (for $n = 34$)	To (for $n = 49$)		From (for $n = 34$)	To (for $n = 49$)			
5.57	0.34	0.44	12	-0.41	-0.35	5	0.77	8
5.57	0.54	0.62	3	-0.21	-0.18	6	1.08	6
5.57	0.71	0.78	3	-0.07	-0.05	1	1.36	2
5.57	0.88	0.94	2	0.05	0.07	4	1.67	8
5.57	1.09	1.13	5	0.18	0.21	2	2.06	5
5.57	1.41	1.44	9	0.35	0.40	7	2.64	4
5.57	5.57	6.78	5	5.57	6.78	14	6.09	6

f = Degrees of freedom.

* $D_1 = \sum (O - E)^2 / E$.

$^\dagger \alpha$ = significance level.

equals zero, is close enough to the observed mean skewness ($\bar{g}_v = 0.15$ for $v = 0$) and so qualifies the distribution as nearly the best among power transformations to normality; the skewness standard deviation $\sigma(G) = 0.36$ for $n = 40$ is appreciably below the observed value $s(g_v) = 0.57$ for $v = 0$ (see also Figure 3). Similar conclusions were also reached by Landwehr *et al.* [1980] from an analysis of U.S. flood data. A chi-square test, carried out to compare the observed distribution of g_v for $v = 0$ with that obtaining under the lognormal law, showed the latter to be unacceptable as a good parent, even at the 1% level (see Table 1). In too many series the upper quantiles are underestimated. The influence of low flows on the observed skewness distribution proved to be negligible.

EXTREME VALUE DISTRIBUTION FOR TWO-COMPONENT MIXTURE VARIABLES

Another possible approach to account for both the presence of outliers and the high skewness of AFS's consists of assuming that flood peaks do not all arise from one and the same distribution, thus allowing for the fact that different types of storms can give rise to different flood characteristics. In the Italian region considered, flood outliers correspond to observed rainfall values much higher than the other annual maxima. Such extremely heavy rainfall is due to extensive storm nuclei engendered by well-developed convective cells in association with special meteorological conditions, as described, e.g., by Fea *et al.* [1968] and Miro-Granada y Gelabert [1974] in connection with catastrophic events which occurred in the Mediterranean region.

Then in modelling AFS's we may assume there are two independent sequences of iid variables Z_{1i} , $i = 1, 2, \dots, K_1$ and Z_{2j} , $j = 1, 2, \dots, K_2$, generated by different types of storm and each defining a compound Poisson process with Poisson parameters given by $\Lambda_1 = E[K_1]$ and $\Lambda_2 = E[K_2]$, respectively, with $\Lambda_1 > \Lambda_2$. The total number $K = K_1 + K_2$ of independent peaks in a year will also be a Poisson process with parameter $\Lambda = \Lambda_1 + \Lambda_2$, while the peak magnitude Z arises from a two-component mixture having CDF

$$F_Z(z) = pF_{Z_1}(z) + (1 - p)F_{Z_2}(z) \quad z \geq 0 \quad (11)$$

where $p = \Lambda_1/\Lambda$ is the proportion of Z_1 in the mixture. One of the two components, the "basic distribution" with CDF $F_{Z_1}(\cdot)$, generates ordinary (more frequent and less severe in the mean) floods. The other, the "outlying distribution," exhibits much greater variability than the basic one and tends to generate rarer but more severe floods.

From (2) and (11) the CDF of the annual flood may be expressed in the form

$$F_X(x) = \exp \{ -\Lambda_1[1 - F_{Z_1}(x)] - \Lambda_2[1 - F_{Z_2}(x)] \} \quad (12)$$

$$x \geq 0$$

or, denoting with X_1 and X_2 the annual maximum of Z_1 and Z_2 , respectively, in the form

$$F_X(x) = F_{X_1}(x)F_{X_2}(x) \quad (13)$$

i.e., as the product of two CDF's, each characterizing the maximum of a compound Poisson process. If Z_1 and Z_2 are both assumed to be exponential random variables, with parameter $\theta_1 = E[Z_1]$ and $\theta_2 = E[Z_2]$ respectively, (13) reduces to

$$F_X(x) = \exp(-\Lambda_1 e^{-x/\theta_1} - \Lambda_2 e^{-x/\theta_2}) \quad x \geq 0 \quad (14)$$

which defines the two-component extreme value (TCEV) distribution. Its four parameters characterize the mean number of independent peaks in a year (Λ_1 and Λ_2) and the mean peak amplitude (θ_1 and θ_2) of the basic and the outlying component. Similarly, as stated in connection with expressions (4) and (4'), CDF (14) may be said to be equivalent to the product of two EV1 CDF's

$$F_X(x) = \exp[-e^{-(x-\varepsilon_1)/\theta_1}] \exp[-e^{-(x-\varepsilon_2)/\theta_2}] \quad (15)$$

with $\varepsilon_j = \theta_j \ln \Lambda_j$ ($j = 1, 2$).

A CDF in the form (15) was first alluded to by Gumbel [1958, p. 198] as the CDF of the larger of two independent EV1-distributed variables for streams with two flood regimes. In a more general form, of the type of (13) but as the product of two asymptotic extreme value CDF's, it was again considered by Canfield and Borgman [1975] and Canfield [1979].

A similar model was suggested by Todorovic and Rousselle [1971] again leading to a CDF of the type of (14) but with seasonally separated components. They used the PDS approach and assumed that only the exceedances which occurred during the same season are identically exponentially distributed and that the exceedances are mutually independent both within and between seasons. From a physical standpoint, however, the assumption of identically distributed flood peaks within a season is not realistic since the separation of floods by season only partially corresponds to separation by meteorological mechanisms. Floods generated from different type storms can "mix" in one season [Gupta *et al.*, 1976]. This is particularly true of the Italian data, for which outliers occur anywhere from August to February, no differently from other floods. In previous papers the TCEV model was applied to PDS's of both daily rainfall [Rossi, 1980; Penta *et al.*, 1980] and daily flow [Versace *et al.*, 1982; Rossi and Versace, 1982]. In all cases when the simple exponential failed, the exceedance magnitude distribution was found to be better modeled by a mixture of two exponential components than by seasonally combined exponential distributions.

A priori subdividing the historic streamflow data into separate subsets is, however, feasible when floods are generated by either rainfall or snowmelt [Waylen and Woo, 1982]. Then two annual maximum series can be obtained, one for each generating process, and fitted by distinct EV1 distributions. The CDF of the annual flood is given by the product (15) of the individual CDF's.

In the Italian region considered, where floods are generated by storms of different type, no such a priori separation of the data is feasible: neither by season, for they can mix in each season, nor by storm characteristics, for the available meteorological information is too scanty. Consequently, the four parameters of the TCEV distribution must be jointly estimated.

PARAMETER ESTIMATION METHODS

Methods for estimating the four parameters of the TCEV distribution depend on the nature of the data to which it is applied (whether PDS's or AFS's are considered). With the former type of data, m independent peak floods exceeding a base level z_0 are considered. The parameters θ_1 , θ_2 , and p of the two-exponential mixture CDF (11) can be evaluated by the ML method [Hasselblad, 1969]. The ML estimator of the Poissonian parameter Λ is simply given, for a 1-year unit of time, by $\hat{\Lambda} = m/n$. The method was used for analyzing some daily flow PDS's of Italian rivers [Versace *et al.*, 1982]. The main difficulty in applying the method is the sad fact that

often, as is the case of Italian rivers, PDS's of instantaneous flows are but rarely available.

When fitting the model to AFS data, the four parameters of the TCEV distribution (13) can be estimated by various methods. Canfield [1979] suggested a numerical procedure based on a least squares technique. There are, however, no real difficulties in using the ML method also here. To this end, an iteration scheme to maximize the likelihood function was developed. The TCEV distribution has a finite probability, $\exp(-\Lambda)$, that the value is exactly zero. Since such a probability is negligible, the TCEV distribution is undistinguishable in practice from a continuous one also allowing for negative values and hence also differentiable for $x = 0$. Then the probability density function of X , $f_X(x)$, may be expressed from (14) in the form

$$f_X(x) = F_X(x)\psi(x) \quad (16)$$

where

$$\psi(x) = (\Lambda_1/\theta_1)e^{-x/\theta_1} + (\Lambda_2/\theta_2)e^{-x/\theta_2} \quad (17)$$

so that the logarithm of the likelihood function, here denoted by L , may be written

$$\begin{aligned} L &= \sum_{i=1}^n \ln f_X(x_i) \\ &= \sum_{i=1}^n \ln F_X(x_i) + \sum_{i=1}^n \ln \psi(x_i) \end{aligned} \quad (18)$$

Equating to zero the first partial derivatives of L with respect to the four parameters to be estimated, we obtain

$$\begin{aligned} \partial L / \partial \Lambda_j &= - \sum_{i=1}^n e^{-x_i/\theta_j} + (1/\theta_j) \sum_{i=1}^n [e^{-x_i/\theta_j} / \psi(x_i)] \\ &= 0 \quad j = 1, 2 \end{aligned} \quad (19)$$

$$\begin{aligned} \partial L / \partial \theta_j &= -(\Lambda_j/\theta_j^2) \left\{ \sum_{i=1}^n x_i e^{-x_i/\theta_j} \right. \\ &\quad \left. + \sum_{i=1}^n [e^{-x_i/\theta_j} (1 - x_i/\theta_j) / \psi(x_i)] \right\} \\ &= 0 \quad j = 1, 2 \end{aligned} \quad (20)$$

Multiplying both sides of (19) by Λ_j and solving (20) for θ_j , we obtain

$$\Lambda_j = \Lambda_j \left[\sum_{i=1}^n e^{-x_i/\theta_j} / \psi(x_i) \right] \left(\theta_j \sum_{i=1}^n e^{-x_i/\theta_j} \right)^{-1} \quad j = 1, 2 \quad (21)$$

$$\begin{aligned} \theta_j &= \left[\sum_{i=1}^n x_i e^{-x_i/\theta_j} / \psi(x_i) \right] \left[\sum_{i=1}^n x_i e^{-x_i/\theta_j} \right. \\ &\quad \left. + \sum_{i=1}^n e^{-x_i/\theta_j} / \psi(x_i) \right]^{-1} \quad j = 1, 2 \end{aligned} \quad (22)$$

The four equations (21) and (22) can be solved by a "successive substitution" iterative scheme. The TCEV distribution was applied to each of the 12 Italian AFS's, which exhibit critically (at 0.05 level) large values of both test statistics $Y_{(n)}$ and G for the EV1 distribution. The parameters were estimated by the ML method. All 12 seemed by eye to fit adequately. In particular, the presence of outliers was accounted for and reconciled. A representative example is illustrated in Figure 5, which shows the TCEV CDF fitted to an n -year AFS with a single observation qualifying as outlier under the simple EV1. Also plotted in the same figure is the theoretical CDF $F_{X_{(n)}}(x) =$

$[F_X(x)]^n$ of the n -year flood $X_{(n)}$. It appears that the largest observed flood is well accounted for, since its exceedance probability in n years is significantly different from zero, whereas it would very closely vanish under an EV1 distribution.

However, when the four parameters of the TCEV distribution are estimated from a single AFS, the uncertainty is great, particularly as regards the parameters of the outlying component. The uncertainty becomes extremely high for AFS's without outliers that give no significantly large values of the appropriate test statistics $Y_{(n)}$ and G . Consequently, a preferable model for quantile estimation may be one using a more parsimonious distribution, e.g., the PEV distribution with $v = 0.5$, unless it is decided to have recourse to a regionalization technique, by which means the uncertainty is reduced thanks to the larger number of data used. Such techniques are of interest for obtaining a reliable estimator of the T -year flood, not only for nongauged but also for gauged sites.

Regional analysis is also necessary for validating the distribution suggested as a parent flood distribution. As is shown in the next section, a regionalized TCEV distribution derived from the set of 39 Italian AFS's is apt to reproduce the empirical distributions of both the skewness coefficient and the reduced largest order statistic.

VALIDATION OF A REGIONALIZED TCEV DISTRIBUTION AS A PARENT FLOOD DISTRIBUTION

A technique of regional analysis was applied by the authors to the analysis of the 39 Italian AFS's. A preliminary consideration before all AFS data of a region may be pooled into a single regional fit is the realization that the parameters θ_1 and Λ_1 of the basic component are hardly constant but vary from site to site within a region. Particularly large are the variations undergone by θ_1 , affected as it is by physiographic and pluviometric factors, but also Λ_1 , although it remains constant

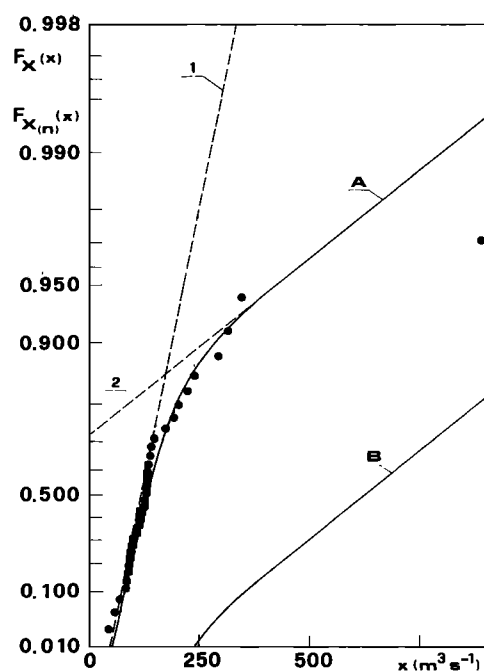


Fig. 5. TCEV distribution fit to Chiani River AFS (curve A). Theoretical CDF's of the basic component annual maximum X_1 (dashed line 1), the outlying component annual maximum X_2 (dashed line 2), and the n -year flood $X_{(n)}$ (curve B).

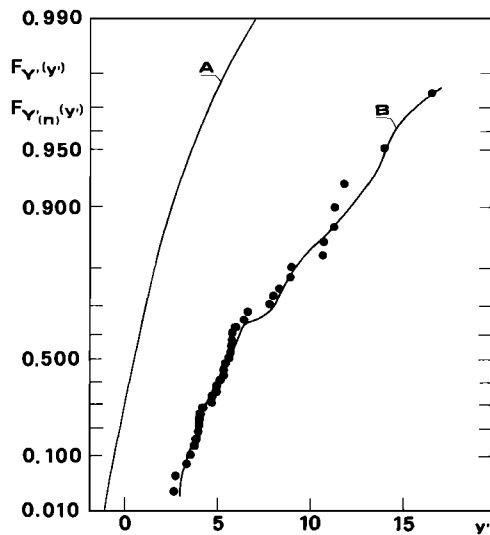


Fig. 6. Regionalized TCEV CDF of the reduced annual flood Y' (curve A). Observed and sampling experiment-derived ($n = 40$) CDF of the reduced n -year $Y_{(n)}'$ (solid circles and curve B, respectively).

within smaller areas of the region, does vary from one subregion to another, essentially in relation to climatic characteristics [Rossi and Versace, 1982]. To allow for such variability of the parameters, a dimensionless variable, the reduced annual flood Y' , was introduced:

$$Y' = (X - \varepsilon_1)/\theta_1 = (X/\theta_1) - \ln \Lambda_1 \quad (23)$$

Its CDF can be obtained from (14) as

$$\begin{aligned} F_{Y'}(y') &= F_X(\theta_1 y' + \varepsilon_1) \\ &= \exp(-\Lambda_1' e^{-y'/\theta_1'} - \Lambda_2' e^{-y'/\theta_2'}) \end{aligned} \quad (24)$$

with parameters

$$\theta_1' = \Lambda_1' = 1 \quad \theta_2' = \theta_2/\theta_1 \quad \Lambda_2' = \Lambda_2/(\Lambda_1^{1/\theta_2'}) \quad (25)$$

Then the simplest hypothesis was introduced, namely, that θ_2' and Λ_2' are regional parameters, remaining constant from site to site. Constancy of θ_2' implies that the ratio θ_2/θ_1 of the mean peak magnitudes of the two components remains constant throughout the region. As regards the Λ_1 and Λ_2 parameters, the implication of constancy of Λ_2' is that the variability of Λ_2 from one subregion to another is much smaller than that of Λ_1 .

It was then considered that the reduced Y' AFS's of all 39 sites are independently originating from one and the same TCEV distribution, namely, (24) with parameters to be estimated, spatial correlation effects being left out of account. For each AFS the parameters θ_1 and Λ_1 were estimated by applying a "rejection of outliers" (RO) procedure, already suggested by Anscombe [1960] as a robust estimation method for samples contaminated by one or more outliers. For each AFS, once its outliers were eliminated by the forward rejection rule, the remaining sample was used for estimating θ_1 and Λ_1 by the ML method on the assumption of an EV1 distribution. Then for each site the reduced annual floods, outliers included, were obtained by substituting the RO-ML estimates $\hat{\theta}_1$ and $\hat{\Lambda}_1$ for θ_1 and Λ_1 in (23).

The 39 reduced AFS's were pooled together to give a single series of 1525 station year data, from which the four parameters of a regionalized TCEV distribution were estimated by the ML method, using the successive substitution procedure.

The distribution of Y' thus obtained, with estimated parameters $\hat{\Lambda}_1' = 0.811$, $\hat{\theta}_1' = 0.867$, $\hat{\Lambda}_2' = 0.161$, and $\hat{\theta}_2' = 2.606$, is plotted in Figure 6. The deviations of the estimates $\hat{\Lambda}_1'$ and $\hat{\theta}_1'$ from the theoretical value one is due to distortion by the RO method.

To validate this regionalized TCEV distribution as an acceptable reduced flood parent distribution, sampling experiments (admittedly limited in scope) were undertaken. Two sets of 40,000 values were independently generated from two EV1 distributions having the above regionalized estimates of the parameters. The larger of each pair of values thus generated was considered, giving 40,000 values of the regionalized TCEV distribution, which were then grouped into 1000 series of length $n = 40$. For each series the skewness coefficient G was calculated from (6) and the basic component parameters were estimated by the rejection of outliers procedure. By using these estimates, the largest value of each series was reduced according to (23). The observed CDF's of the reduced largest order statistic $Y_{(n)}'$ and of the skewness coefficient G were compared with the sampling experiment CDF's of Figures 6 and 7, respectively. The comparison shows that the regionalized TCEV distribution reproduces the observed distribution of both G and $Y_{(n)}'$ extremely well.

As it was for the PEV distribution, the chi-square test was also applied to check the goodness of fit to the observed distribution of G , but in this case the variability of the CDF of G with n was ignored, and n was set equal to 40 throughout. The number of degrees of freedom was identified with the number of classes minus three, paying tribute to the fact that the skewness G is dependent on parameters θ_2' and Λ_2' regionally estimated, whereas it is independent of the parameters of the basic component. Table 1 shows that the test statistic is exceeded with 29% probability, much higher than the 1.4% probability obtained for the PEV distribution with $\nu = 0.5$.

Thus the regionalized TCEV distribution satisfactorily accounts for the condition of separation and the presence of outliers, even ignoring possible spatial mixing of parameters θ_2' and Λ_2' . It may be deemed acceptable as a good parent distribution and preferable to the PEV distribution (based on identically distributed floods).

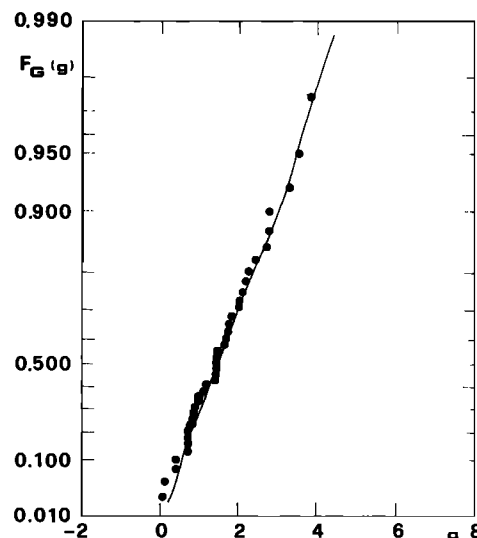


Fig. 7. Observed and sampling experiment-derived ($n = 40$) CDF of the skew G .

CONCLUSION AND DISCUSSION

The main purpose of the investigation reported was the identification and validation of a "good" parent flood distribution having parameters representative of actual flood records. The type of distribution selected was in keeping with the physical basis of the flood process, the annual flood being identified with the largest value of a compound Poisson process. Among distributions of this type the most suitable was sought on the basis of an analysis of the statistical characteristics of a set of 39 AFS's of Italian basins. The high variability exhibited by the skewness coefficient G of these data was attributed to the random presence of isolated flood values much higher than the rest, values that qualify as outliers under the EV1 distribution. More heavily tailed distributions obtained by power law transformations also failed to account adequately for the observed distribution of G .

To overcome this deficiency the two-component extreme value (TCEV) distribution was here suggested. It assumes the individual floods to arise from a mixture of two exponential distributions, referred to as the "basic" and the "outlying" component. The corresponding CDF of the annual flood may also be interpreted as the CDF of the larger of two EV1 distributed random variables. This four-parameter distribution emerges as a generalization of the EV1 distribution to which it reduces in default of the outlying component. The TCEV parameters may be estimated from either PDS or AFS series by the ML method.

By using a regionalization technique, a regional TCEV distribution was derived with parameters representative of the set of 39 Italian AFS's. For each station, flood records were depurated of the outliers and reduced, by a linear transformation, to normalize the parameters of the basic component. The reduced data were pooled into a single series from which regionalized parameters were estimated. By limited sampling experiments, the regionalized TCEV distribution obtained was recognized to reproduce closely the observed distribution of both the skewness and the largest order statistic. It can accordingly be accepted as a good parent flood distribution. Some issues deserving further investigation remain.

A natural candidate for further probing is the hypothesis that the variations of the outlying component parameters θ_2 and Λ_2 from one basin to another are so related to the corresponding variations of the basic component parameters θ_1 and Λ_1 as to leave constant the regional parameters θ_2' and Λ_2' defined by (25). So far we were unable to identify any clear-cut variation in θ_2' and Λ_2' from site to site that could be related to corresponding variations in the site climatic and geomorphic characteristics, nor were we able to subdivide the region into subregions with distinct values of θ_2' and Λ_2' . The specific information provided by at-site or at-subregion data, although adequate for recognizing the variability of θ_1 and Λ_1 from site to site, is not so for θ_2' and Λ_2' . Further information from climatological and hydrological analyses of outlying flood events may be helpful. A systematic analysis of the available series of daily rainfall annual maxima (which, besides, are less affected by measuring error than is the case of discharge data) should also prove useful. Such investigations may promote the birth of a better parent flood distribution.

Deserving further thought also is the use of the regionalized TCEV distribution as a basis for quantile estimation in preference to models more parsimonious in the number of parameters, such as the two-parameter PEV distribution with $v = 0.5$. To clarify this issue requires evaluating the sampling proper-

ties of the estimation procedure suggested, which could be carried out by Monte Carlo techniques incorporating reasonable interstation correlation between concurrent flows.

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